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PAPER_1000: NS Merger F_U_Bi with Strain Suppression

Abstract

We extend the NS merger F_U_Bi framework (GW190425) with 3rd-order Ramanujan $S_{26}^{(3)}$, incorporating BCS gap coupling and tidal correction. At resonance, 47.0% strain reduction is achieved: h_UQFF = h_GR · (1 − 0.47) = 0.53 · h_GR.

1. Strain Suppression

$h\_UQFF(\Gamma) = h\_GR \cdot (1 - 0.47 \cdot \Phi(\Gamma)/S_{26}^{(3)})$

For GW190425 (M_total = 3.4 M☉, d = 159 Mpc): h_GR = 2.52×10^{-22} → h_UQFF = 1.33×10^{-22}.

2. Mass-Gap Classification

m_1 = 2.52 M☉ → P(NS) = 49%, P(BH) = 51%. Phonon suppression factor discriminates mass-gap objects.

3. Implementation

File: `fubi_agn_ns_mergers.py`, class `NSMergerFUBiCalc`. CP4 class #584.